

GoopOnTheGo

Team Project Report

CSD3156 Mobile and Cloud Computing
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GitHub Repository <https://github.com/SquigglyOwl/GoopOnTheGo>

App Demo Video

Presentation Video

Team Contributions

Member	Contributions
Edwin	Core game architecture, AR camera pipeline and colour detection CV algorithm, catch mechanic, evolution and fusion systems, Room database design and seeding, GPS habitat mapping, daily challenges, achievement system, login streak, Goopdex with persistent discovery, achievements header button, CI/CD pipeline, unit tests, Report, Video
Jeremy	UI overhaul - gradient header drawable, vector nav icons (scan, collection, map, back), card elevation and item layout polish across all screens, colour palette in colors.xml, light/night theme tokens; released all duplicates feature (v1.0) with DAO query and repository logic; moved release button from EvolutionActivity to CollectionActivity for better accessibility, Report
Kai Rui	Confirmation dialog before releasing a creature (with 3-second cooldown timer on the release button); display of caught location on CreatureDetailActivity (shows GPS coordinates or habitat zone name if GPS was unavailable), Report
Nur Aqif	Search bar on Collection screen - TextInputEditText with real-time Text Watcher filtering creatures by name and nickname (case-insensitive); sort dropdown positioned below search bar, Report
Ridhwan	Live creatures count on the Collection button in HomeActivity (e.g. "Collection (5)"); sort Collection dropdown (by name, rarity, date caught), Report

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1. Introduction & Game Overview

GoopOnTheGo is a location-based creature collection game for Android, inspired by Pokemon Go. Players scan their physical environment using the device camera to spawn and catch creatures called “Goops.” Each creature type is linked to a real-world colour signature detected via the camera in real time. Players build a collection, evolve creatures by merging duplicates, fuse two different types into hybrid creatures, and track progress through achievements and daily challenges.

Core Game Loop

Camera Scan → Colour Detected → Creature Spawns → Tap to Catch → Collect → Evolve / Fuse

Creature System

- **19 creatures** (15 + 4) across 5 base types (Water, Fire, Nature, Electric, Shadow) and 4 hybrid fusion types
- **3-stage evolution chains** - e.g. Droplet Goop → Aqua Goop → Tsunami Goop
- **4 fusion combinations** producing hybrid creatures - e.g. Water + Fire = Steam Goop

Key Features Summary

Feature	Description
AR-style Camera Scanning	Real-time colour detection triggers creature spawns
Catch Mechanic	Tap creature within radius; rarity determines success rate
Creature Collection	Filterable inventory with nicknames and favourites
Evolution	Collect 3 identical creatures and merge them to evolve from either the Evolution screen or the Creature Detail screen
Fusion	Combine 2 compatible types into a hybrid creature
Habitat Map	GPS-based map showing nearby creature habitat zones

Feature	Description
Daily Challenges	Time-limited objectives refreshed each midnight
Achievements	10 persistent achievements across 5 categories
Login Streaks	Consecutive daily login tracking

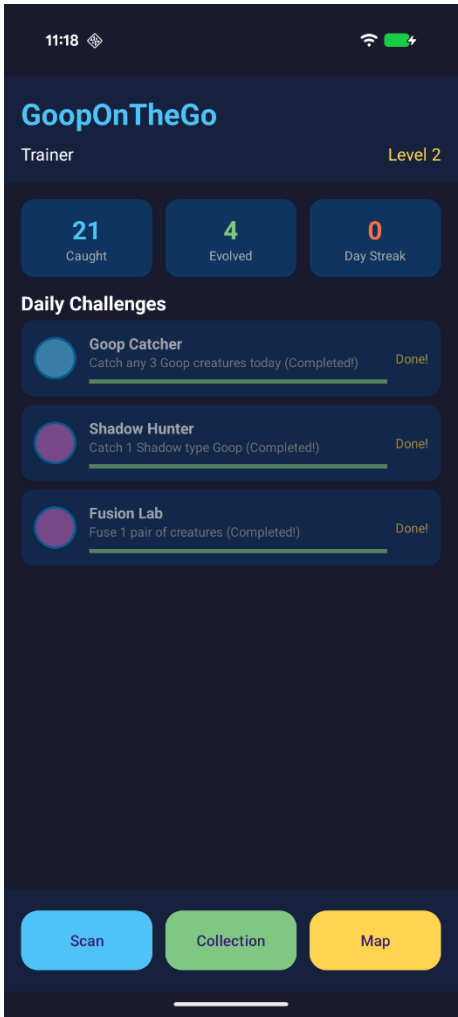
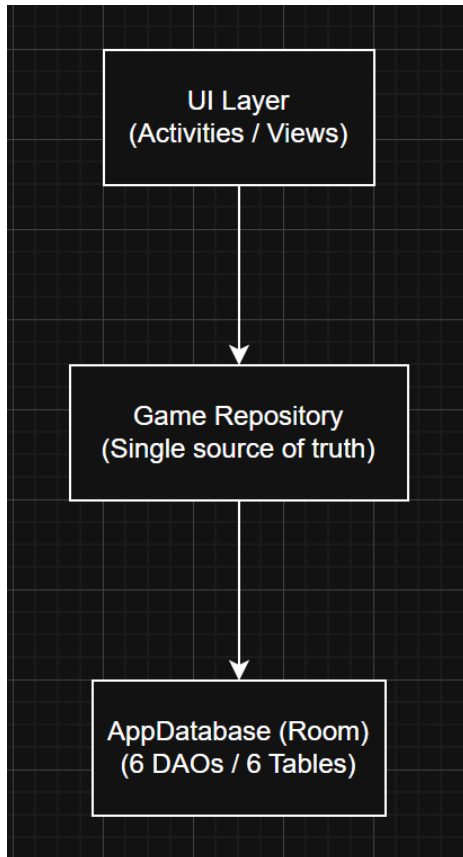


Figure 1: Home screen showing player stats and daily challenges

2. Software Architecture

The app follows an **MVVM-adjacent architecture** using the Repository pattern with Android Jetpack components.

Architecture Diagram



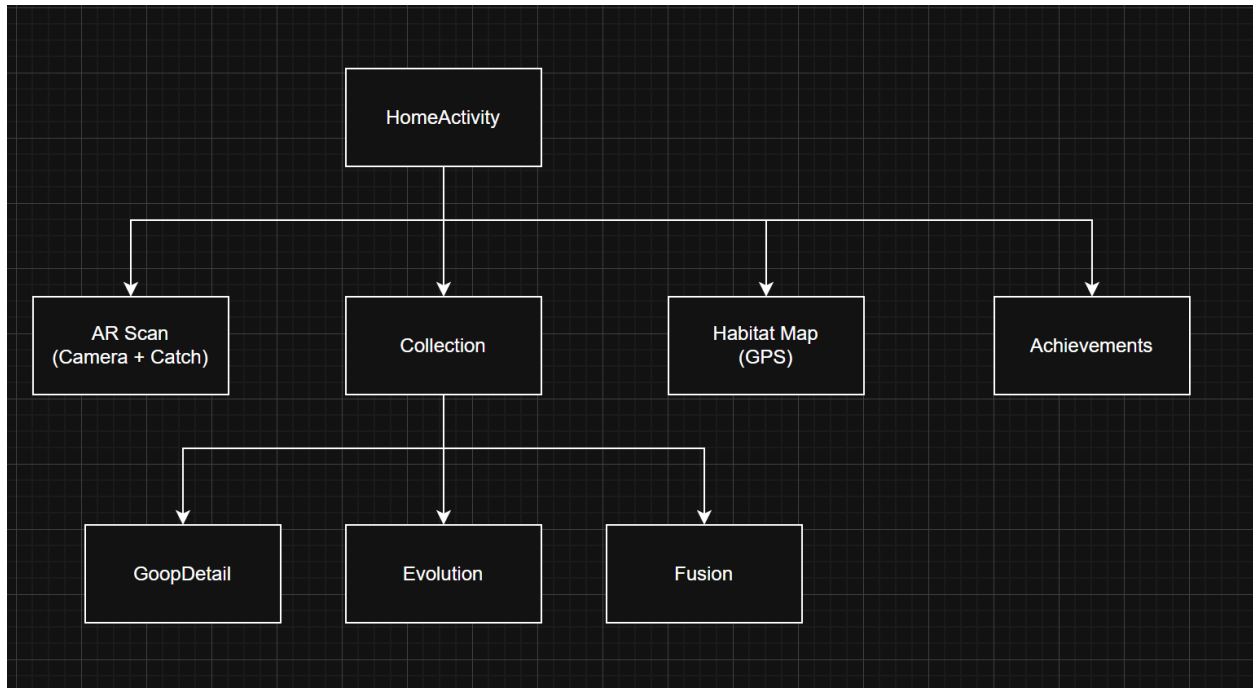
Architectural Decisions

Pattern	Implementation	Rationale
Repository Pattern	GameRepository.kt - centralises all data access and game logic	Keeps the UI separate from the data layer
Reactive Data	Kotlin Flow $\Leftarrow$ from Room DAOs	UI updates automatically when data changes, no

Pattern	Implementation	Rationale
		polling needed
Coroutines	lifecycleScope.launch, Dispatchers.IO / Main	Database and camera work runs off the main thread
Singleton App	GoopApplication holds database and repository lazily	Shared across all screens
Custom Views	AROverlayView, HabitatMapView	Canvas drawing without a game engine

Application Entry: GoopApplication.kt initialises the Room database and GameRepository lazily, making them available to all Activities via (application as GoopApplication).repository.

Navigation Flow



Activities Overview

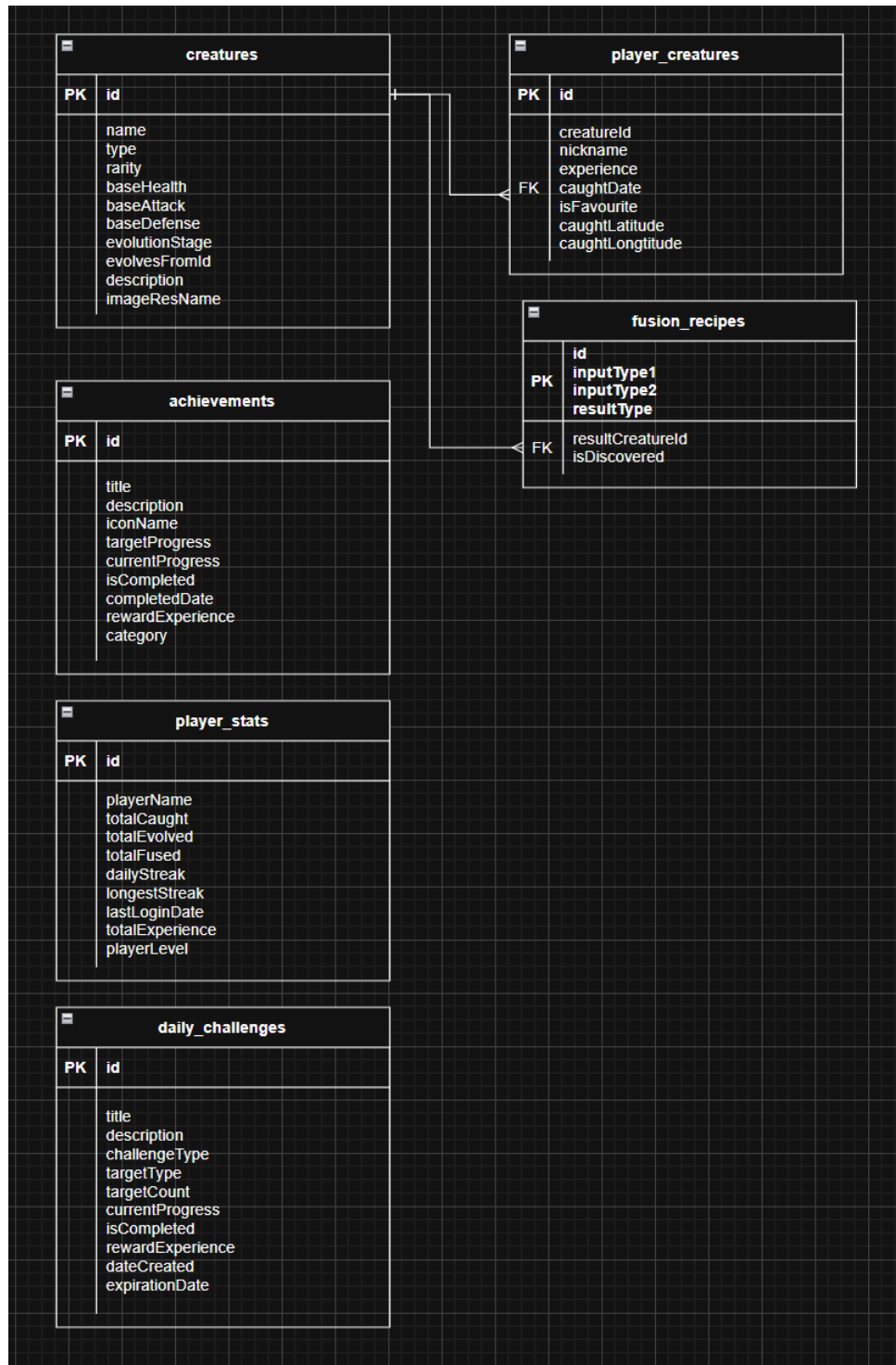
Activity	Purpose
HomeActivity	Main hub - player stats, daily challenges, navigation
ARScanActivity	Camera preview, colour analysis, creature spawning, catch mechanic
CollectionActivity	RecyclerView grid (3 columns), type and favourite filtering
CreatureDetailActivity	Creature stats, nickname editing, evolve/release actions
EvolutionActivity	Lists creatures with 3+ copies ready to evolve; shows creature images in evolution preview card

Activity	Purpose
FusionActivity	Two-slot selection for type fusion
HabitatMapActivity	Live GPS tracking, habitat zone detection
AchievementsActivity	Achievement list with progress bars
GoopdexActivity	Creature encyclopedia - silhouette for undiscovered, full image for discovered; live discovery counter (e.g. “7 / 19”); accessible from Collection screen

3. Database Design

The app uses Room ORM on top of SQLite with a database named gooponthego_database (schema version 2), with 6 entity tables.

3.1 Entity-Relationship Overview



3.2 Table Definitions

creatures - Master creature list (19 rows seeded on first launch)

Column	Type	Constraints	Purpose
id	LONG	PK, autoGenerate	Unique creature ID
name	STRING	NOT NULL	Display name
type	STRING	NOT NULL	GoopType enum via TypeConverter
rarity	INT	NOT NULL	1–5 (1 = common, 5 = legendary)
baseHealth	INT	NOT NULL	Base HP stat
baseAttack	INT	NOT NULL	Base attack
baseDefense	INT	NOT NULL	Base defense
evolutionStage	INT	NOT NULL	1, 2, or 3
evolvesFromId	LONG	NULLABLE	Reference to parent form
evolvesToId	LONG	NULLABLE	Reference to next stage
description	STRING	NOT NULL	Dex entry text
imageResName	STRING	NULLABLE	Drawable resource name

player_creatures - Player's caught creature instances

Column	Type	Constraints	Purpose
id	LONG	PK, autoGenerate	Unique instance ID
creatureId	LONG	FK → creatures.id, CASCADE DELETE	Base creature reference
nickname	STRING	NULLABLE	Player-assigned name
experience	INT	DEFAULT 0	Accumulated XP
caughtDate	LONG	DEFAULT now	Epoch timestamp of capture
isFavorite	BOOLEAN	DEFAULT false	Favourite flag

Column	Type	Constraints	Purpose
caughtLatitude	DOUBLE	NULLABLE	GPS latitude at capture
caughtLongitude	DOUBLE	NULLABLE	GPS longitude at capture

player_stats - Singleton player progress (always id = 1)

Column	Type	Purpose
totalCaught / Evolved / Fused	INT	Cumulative event counters
dailyStreak / longestStreak	INT	Login streak tracking
lastLoginDate	LONG	For streak day-diff calculation
totalExperience	INT	Account-level XP
playerLevel	INT	Derived: (totalExperience / 1000) + 1

daily_challenges - Time-limited objectives (expire at midnight)

Column	Type	Purpose
challengeType	STRING (Enum)	CATCH_ANY, CATCH_TYPE, EVOLVE, FUSE, SCAN_LOCATIONS
targetType	STRING?	GoopType for type-specific challenges
targetCount / currentProgress	INT	Goal vs current progress
expirationDate	LONG	Next day 00:00:00 epoch
rewardExperience	INT	XP on completion

achievements - Persistent unlock system (10 seeded achievements)

Column	Type	Purpose
category	STRING (Enum)	COLLECTION, EVOLUTION, EXPLORATION, DAILY,

Column	Type	Purpose
		SPECIAL
targetProgress / currentProgress	INT	Goal tracking
completedDate	LONG?	Null until completed
rewardExperience	INT	XP reward

fusion_recipes - 4 fusion combination rules

Column	Type	Purpose
inputType1 / inputType2	STRING	GoopType ingredients
resultType	STRING	Output hybrid type
resultCreatureId	LONG	FK to resulting creature
isDiscovered	BOOLEAN	Revealed to player on first use

3.3 Fusion Recipes

Type 1	Type 2	Result
WATER	FIRE	STEAM
ELECTRIC	NATURE	LIGHTNING_PLANT
FIRE	NATURE	MAGMA
WATER	ELECTRIC	ICE

3.4 Notable Queries

Bidirectional Fusion Recipe Lookup (FusionRecipeDao):

```
SELECT * FROM fusion_recipes
WHERE (inputType1 = :type1 AND inputType2 = :type2)
OR (inputType1 = :type2 AND inputType2 = :type1)
```

This matches a recipe regardless of the order the two creature types are passed in.

Joined Collection Query (PlayerCreatureDao):

```
SELECT * FROM player_creatures  
INNER JOIN creatures ON player_creatures.creatureId = creatures.id
```

Returns all player creatures with their full base creature data in a single transaction.

3.5 Type Conversion

Converters.kt provides Room with `@TypeConverter` functions to persist the `GoopType` enum as a String in SQLite and restore it on read.

4. Mobile Features Implemented

The app uses six advanced mobile features, more than the required minimum of three.

4.1 Camera / Computer Vision

Classes: ARScanActivity.kt, inner class ColorAnalyzer

CameraX ImageAnalysis with a single-thread ExecutorService processes camera frames continuously. A custom pixel-sampling and colour-classification algorithm detects the dominant environmental colour and maps it to a `GoopType`. A 2-second temporal hold filter prevents accidental spawns from transient colour changes. *(Full algorithm documented in Section 6 - Bonus Component.)*

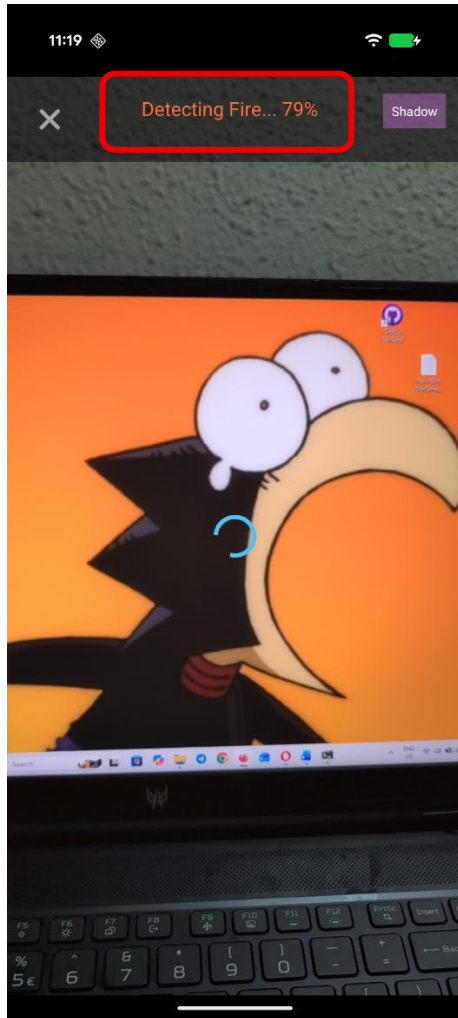


Figure 2: AR scanning screen detecting environmental colour - 2-second temporal hold timer in progress before creature spawn

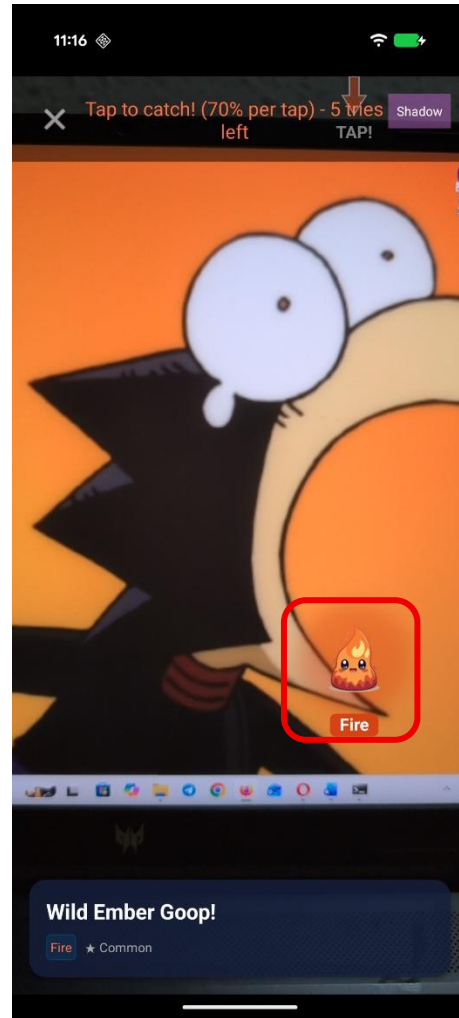


Figure 3: AR scanning screen showing a creature spawned via real-time colour detection

4.2 GPS Location Services

Class: HabitatMapActivity.kt

The **Fused Location Provider** (com.google.android.gms:play-services-location) tracks the player's position with update interval of 5 seconds and minimum update interval of 2 seconds.

Permissions: ACCESS_FINE_LOCATION and ACCESS_COARSE_LOCATION. GPS coordinates

feed into the geohashing algorithm to determine the local habitat type, displayed on the custom canvas map.

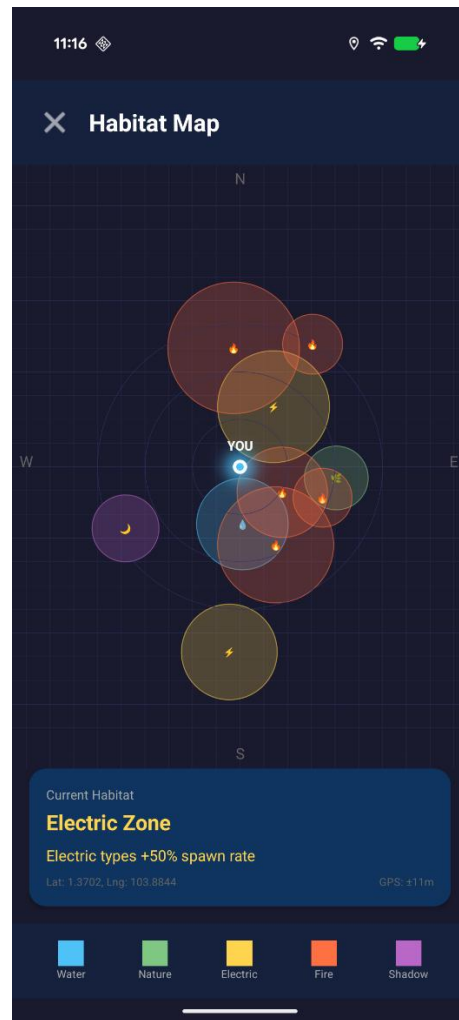


Figure 4: Habitat map showing GPS-tracked player position and habitat zones

4.3 SQLite Database (Room ORM)

Classes: AppDatabase.kt, 6 DAOs, GameRepository.kt

A full relational database with **6 tables**, **foreign key constraints**, **cascade deletion**, **type converters**, and **23 queries** across 6 DAOs. The database is auto-initialised with 19 creatures, 4 fusion recipes, and 10 achievements via RoomDatabase.Callback.

4.4 Canvas-Based 2D Graphics

Classes: AROverlayView.kt, HabitatMapView.kt

Two custom View subclasses render game graphics entirely via the Android **Canvas API** - no external game engine is used. AROverlayView renders animated creatures with procedural blob shapes, layered glow effects, and a directional catch arrow. HabitatMapView renders a live map with habitat zones, a player position marker, a grid background, and a compass.

4.5 Animations

Class: AROverlayView.kt

Two ValueAnimator instances drive continuous creature animations:

- **Bounce:** 1000 ms cycle, AccelerateDecelerateInterpolator, plus/minus 25 px vertical displacement, INFINITE / REVERSE repeat
- **Glow pulse:** 1500 ms cycle, 0–30 px BlurMaskFilter radius expansion, INFINITE / REVERSE repeat

Both animators call invalidate() on each frame to trigger canvas redraw.

4.6 Multi-Threading / Coroutines

All database operations run on Dispatchers.IO via Kotlin coroutines (lifecycleScope.launch). Camera frame analysis runs on a dedicated ExecutorService. Results are dispatched back to Dispatchers.Main for UI updates. The main thread is never blocked.

5. Technical Implementation Details

5.1 Evolution System

Evolution uses a merge-to-evolve mechanic. The player must own 3 identical base-stage creatures. All 3 copies are removed and one evolved creature is inserted. The evolved creature inherits the nickname of the original.

- Requirement: `sameCreatures.size >= 3`
- Action: `delete instances[0..2], insert PlayerCreature(evolution.id)`
- XP reward: 50 XP added to `player_stats`

Evolution can be triggered from two places in the app:

1. `CreatureDetailActivity` - when viewing a specific Goop, a progress bar shows X/3 copies owned. The Evolve button enables automatically once 3 copies are collected.
2. `EvolutionActivity` - a dedicated screen accessible from the Collection screen. It displays all creature types where the player owns 3 or more copies, showing the actual creature image alongside the evolved form in a preview card. Previously this screen used an XP-based filter which never surfaced any creatures; this was identified and fixed so the screen now correctly reflects the merge-to-evolve mechanic.

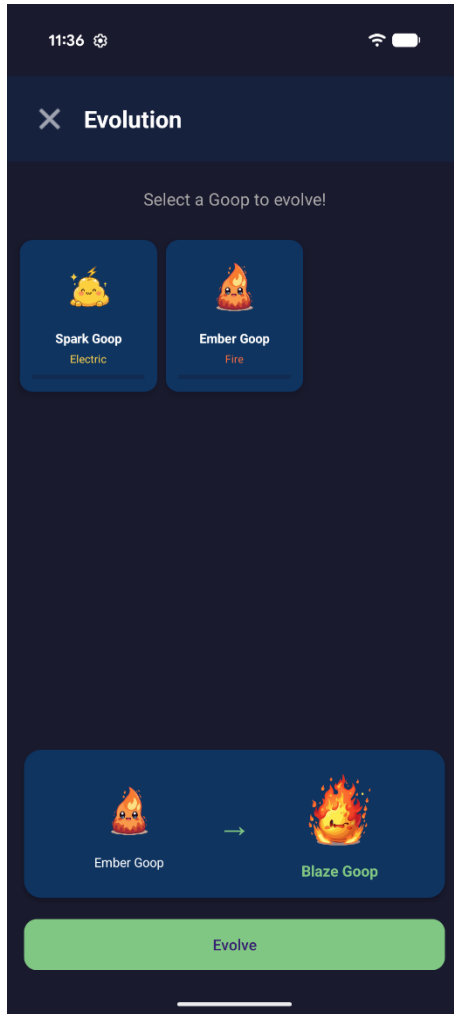


Figure 5: EvolutionActivity showing eligible creatures and evolution preview with creature images

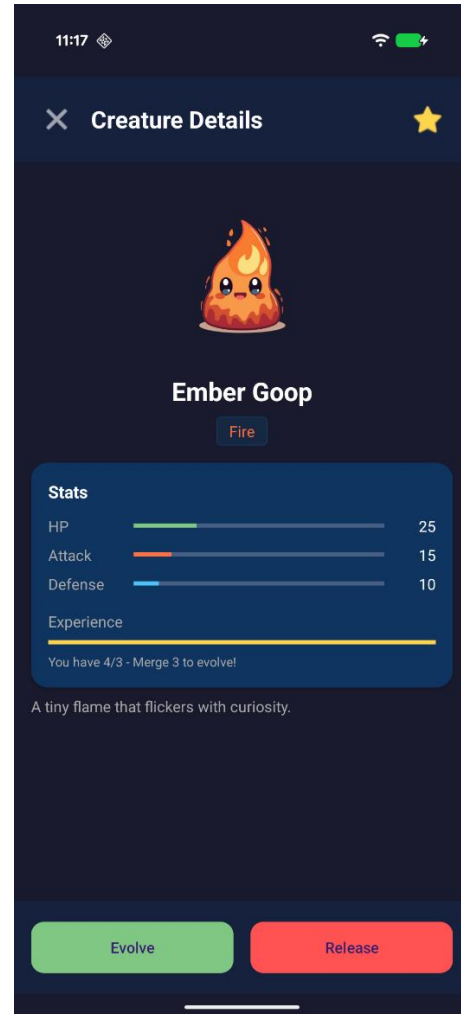


Figure 6: CreatureDetailActivity displaying X/3 evolution progress bar with Evolve button enabled

5.2 Fusion System

Two creatures of compatible types are selected by the player. FusionRecipeDao performs a bidirectional lookup for a matching recipe. Both source creatures are deleted and one hybrid creature is inserted with averaged experience. The recipe is flagged `isDiscovered = true` on first successful use.

- Fused XP = $(\text{experience1} + \text{experience2}) / 2$
- XP reward: 75 XP added to player_stats

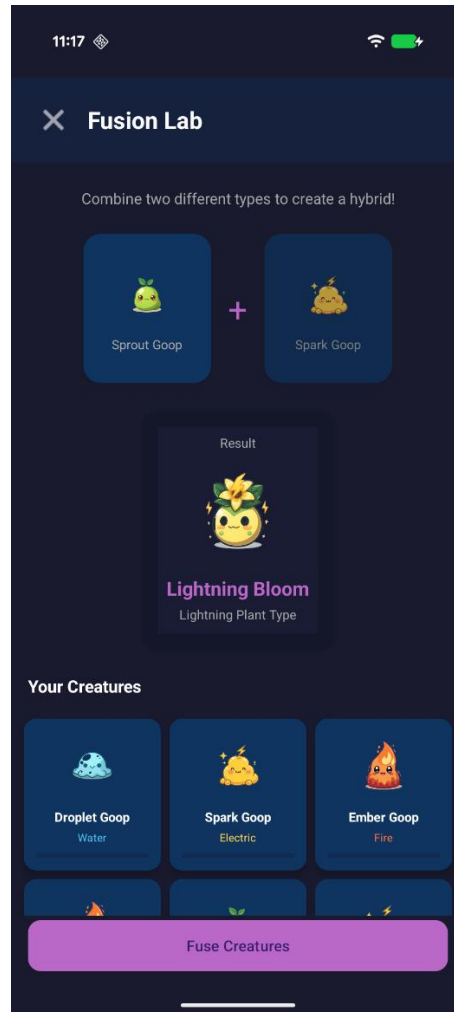


Figure 7: FusionActivity showing two compatible creature slots selected for fusion

5.3 Daily Challenge System

Challenges are generated fresh when no active (non-expired) challenges exist. Three challenges are created with expiration set to the next day at 00:00:00. Progress is updated by event hooks inside GameRepository that fire on catch / evolve / fuse actions.

Default Daily Challenges:

Title	Type	Target	Reward
Catch 3 Goops	CATCH_ANY	3	50 XP
Water Hunter	CATCH_TYPE (WATER)	2	75 XP
Evolution Time	EVOLVE	1	100 XP

5.4 Daily Login Streak

The streak system compares Calendar.DAY_OF_YEAR and Calendar.YEAR between the last login timestamp and the current time:

- Same day: streak unchanged
- Next consecutive day: streak incremented by 1
- Gap of 2+ days: streak resets to 1
- Cross-year transitions handled by comparing both year and day-of-year fields

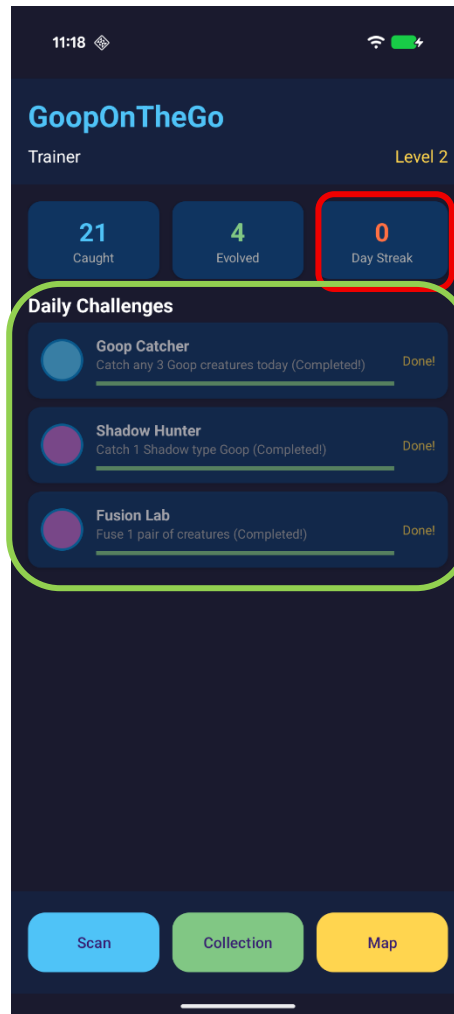


Figure 8: Home screen zoomed in on the *daily challenges* and *daily login* sections

5.5 Achievement System

10 achievements across 5 categories (COLLECTION, EVOLUTION, EXPLORATION, DAILY, SPECIAL). Progress is updated passively on game events and compared against targetProgress. Once isCompleted = true, progress stops updating. Completion awards rewardExperience XP.

Title	Category	Target	Reward
First Catch	COLLECTION	1 catch	50 XP

Title	Category	Target	Reward
Collector	COLLECTION	10 catches	100 XP
Goop Master	COLLECTION	50 catches	500 XP
First Evolution	EVOLUTION	1 evolution	100 XP
Evolution Expert	EVOLUTION	10 evolutions	300 XP
Mad Scientist	EVOLUTION	1 fusion	150 XP
Explorer	EXPLORATION	5 locations	200 XP
Dedicated Trainer	DAILY	7-day streak	250 XP
Type Specialist	COLLECTION	5 types caught	300 XP
Challenge Champion	DAILY	10 challenges	200 XP

5.6 Probabilistic Catch Mechanic

Tap detection computes Euclidean distance from the touch point to the creature's current animated position (accounting for live bounce offset). A tap within 90% of the creature radius triggers a catch attempt:

Rarity	Stars	Catch Rate
Common	1 star	70%
Uncommon	2 stars	55%
Rare	3 stars	40%
Epic	4 stars	25%
Legendary	5 stars	15%

`Random.nextFloat() < catchRate` determines success or failure.

Players have a maximum of 3 attempts before the creature escapes, making each throw meaningful.

5.7 Creature Type System - GoopType.kt

A Kotlin enum with 10 values, each carrying `displayName`, `primaryColor`, `secondaryColor`, and `habitat`. A companion object implements `getFusionResult(type1, type2)` using `setOf()` for order-independent matching.

Type	Primary Colour	Habitat
WATER	Blue (#4FC3F7)	Near water sources
FIRE	Red-orange (#FF7043)	Warm areas
NATURE	Green (#81C784)	Parks and plants
ELECTRIC	Yellow (#FFD54F)	Near electronics
SHADOW	Purple (#BA68C8)	Dark areas
STEAM	Grey (#B0BEC5)	Misty areas
LIGHTNING_PLANT	Yellow-green (#AED581)	Stormy gardens
MAGMA	Deep orange (#FF5722)	Volcanic regions
ICE	Light blue (#E1F5FE)	Cold areas
VOID	Dark indigo (#311B92)	Unknown

5.8 Custom AR Overlay Graphics - AROverlayView.kt

Procedural Blob Rendering uses a 12-point polygon with sine-wave radial distortion:

```
val angle = (i.toFloat() / 12) * 2 * PI
val wobble = sin(angle * 3 + System.currentTimeMillis() / 200.0) * 15f
val r = radius + wobble.toFloat()
val x = cx + (r * cos(angle)).toFloat()
val y = cy + (r * sin(angle)).toFloat()
```

- Wobble cycles at approximately 5 Hz (time / 200)
- 3rd-order angular frequency (angle $\times$ 3) creates 3 lobes of organic distortion
- Rendered with a drop shadow, primary filled path, and white highlight circle

- Glow effect uses `BlurMaskFilter(glowRadius + 20f, BlurMaskFilter.Blur.NORMAL)` drawn behind the creature

5.9 Custom Habitat Map - `HabitatMapView.kt`

Habitat Zone Generation uses random polar coordinates:

```
val angle = random.nextDouble() * 2 * PI
val distance = 100f + random.nextFloat() * 300f // 100–400 px from centre
val offsetX = (cos(angle) * distance).toFloat()
val offsetY = (sin(angle) * distance).toFloat()
val radius = 60f + random.nextFloat() * 80f // 60–140 px zone radius
```

8–12 zones are generated per session. Canvas layers drawn in order: dark grid background, habitat circles, range rings (100/200/300 px), player glow marker, compass labels.

5.10 Geohashing for Habitat Zones - `HabitatMapActivity.kt`

```
val latZone = (lat * 100).toInt() // 0.01 degree resolution = approx 1.1 km
val lngZone = (lng * 100).toInt()
val hash = abs(latZone + lngZone)

return when (hash % 10) {
    0, 1 -> GoopType.WATER
    2, 3 -> GoopType.NATURE
    4, 5 -> GoopType.ELECTRIC
    6, 7 -> GoopType.FIRE
    else -> GoopType.SHADOW
}
```

5.11 Goopdex - Persistent Discovery System

The Goopdex (`GoopdexActivity`) shows all 19 creatures in a 3-column grid. Undiscovered creatures display a black silhouette with “???” for name and type. Discovered creatures show their full colour image, name, and type.

Discovery is saved as `isDiscovered` on the Creature entity - a flag that stays true once set. It gets set to true by `GameRepository` in three cases:

Event	Action
catchCreature()	Marks the caught creature's base entry as discovered
evolveCreature()	Marks the evolved-stage creature as discovered when created
fuseCreatures()	Marks the fusion result creature as discovered when created

This means if a player catches Droplet Goop, evolves all 3 copies into Aqua Goop, and then fuses Aqua Goop away - Droplet Goop, Aqua Goop, and Steam Goop all remain shown in the Goopdex permanently, even though none are currently in the collection.

The silhouette effect is applied via a ColorMatrix that sets all RGB channels to zero while keeping the alpha channel, so the original image shape fills in as solid black.

The Goopdex is accessible from the Collection screen via the “Goopdex” button, and the header shows a live progress counter (e.g. “7 / 19”).

The hash is deterministic - the same real-world location always produces the same habitat type, so the same real-world area always maps to the same habitat type.

6. Bonus: Computer Vision - Real-Time Colour-Based AR Spawning

This section describes the custom CV component we built for the bonus criterion: “Complex implemented component with special algorithms, or other advanced features e.g. CG, ML, CV, VR, AR (mostly using a library or copying code from somewhere doesn’t count). This component should be clearly documented within the report.”

Overview

ARScanActivity.kt implements a real-time computer vision pipeline using CameraX ImageAnalysis to classify the player’s physical environment by dominant colour and map it to a creature type. The pixel sampling, saturation filtering, type classification rules, and temporal hold detection are all **custom-implemented logic** - no CV or ML library is used for the classification itself.

Pipeline Architecture

```
Camera Frame (CameraX ImageAnalysis, every 200 ms)
|
v
Bitmap Capture
|
v
Dominant Colour Extraction <-- Custom sparse pixel sampling
|
v
Saturation Filter <-- Reject achromatic surfaces
|
v
RGB Channel Classification <-- Map colour to GoopType
|
v
Temporal Hold Timer <-- 2-second sustained colour required
|
v
Creature Spawn Trigger <-- evolutionStage = 1 creatures only
```

Step 1 - Dominant Colour Extraction

The centre region of each camera frame is sampled. Every 4th pixel in both axes is read (1/16th of the region) to balance accuracy with CPU performance on the camera thread:

```
for (x in startX..endX step 4) {
  for (y in startY..endY step 4) {
    val pixel = bitmap.getPixel(x, y)
    rSum += Color.red(pixel)
    gSum += Color.green(pixel)
    bSum += Color.blue(pixel)
    count++
  }
}
```

```
}  
return Color.rgb(rSum / count, gSum / count, bSum / count)
```

Step 2 - Saturation Filter

Pixels with low RGB spread are rejected as achromatic. This prevents white walls, grey floors, or overcast skies from falsely triggering a spawn:

```
val max = maxOf(r, g, b)  
val min = minOf(r, g, b)  
if (max - min < 30) return null // Too grey / neutral - no type
```

Step 3 - RGB Channel Type Classification

```
return when {  
    b > r && b > g && b > 80 -> GoopType.WATER // Blue dominant  
    r > g && r > b && r > 80 -> GoopType.FIRE // Red dominant  
    g > r && g > b && g > 80 -> GoopType.NATURE // Green dominant  
    r > 120 && g > 120 && b < 100 -> GoopType.ELECTRIC // Yellow: R+G high, B low  
    r < 50 && g < 50 && b < 50 -> GoopType.SHADOW // Near-black  
    else -> null  
}
```

The ELECTRIC case is the most notable: yellow is not a primary additive colour, so it is identified by requiring both red and green channels to exceed 120 with blue below 100 - a deliberate multi-channel rule rather than single-channel dominance.

Step 4 - Temporal Hold Timer

A single camera frame is unreliable due to lighting flicker and motion. A 2-second sustained colour hold is required before any spawn is triggered. If the detected type changes mid-scan, the timer resets, stopping false spawns when the detected colour changes briefly:

```
private var currentDetectedType: GoopType? = null  
private var colorDetectionStartTime = 0L  
private val COLOR_HOLD_TIME_MS = 2000L
```

```

if (detectedType == currentDetectedType) {
    val elapsed = currentTimeMillis - colorDetectionStartTime
    val progress = ((elapsed / 2000.0) * 100).toInt().coerceIn(0, 100)
    // UI feedback: "Detecting Water... 47%"
    if (elapsed >= COLOR_HOLD_TIME_MS) spawnCreatureOfType(detectedType)
} else {
    currentDetectedType = detectedType
    colorDetectionStartTime = currentTimeMillis // Reset on type change
}

```

Step 5 - Spawn Selection

Only base-stage creatures (evolutionStage = 1) can be wild-spawned via colour detection. Evolved and hybrid creatures are exclusive rewards for gameplay investment (merging, fusing):

```

SELECT * FROM creatures WHERE type = :type AND evolutionStage = 1

```

CV Summary

Aspect	Detail
Input	CameraX ImageAnalysis bitmap frames at 200 ms intervals
Sampling	Sparse centre-region sampling (1/16 pixel density)
Saturation filter	$\max(R,G,B) - \min(R,G,B) < 30$ rejects achromatic frames
Classification	5-rule RGB channel comparison mapping to 5 GoopTypes
Temporal filter	2-second same-type hold with real-time UI progress feedback
Output	spawnCreatureOfType(GoopType) → DB query → AR overlay render
Libraries used	CameraX for frame delivery only -

Aspect	Detail
	classification is entirely custom code

7. Software Engineering Practices

7.1 Version Control

- 15 pull requests merged into master across branches: edwin, jeremy, kairui, aqif, Ridhwan, feature/unit-tests, report
- Feature branch workflow - each team member worked on a named branch and submitted a PR for review
- Commit history shows the project was built up step by step

7.2 CI/CD - GitHub Actions

A GitHub Actions workflow (`.github/workflows/android-ci.yml`) runs automatically on every push and pull request to master, main, and develop:

Step	Action
1	Checkout code - <code>actions/checkout@v4</code>
2	Set up JDK 17 (Temurin distribution)
3	Configure Gradle - <code>gradle/actions/setup-gradle@v4</code>
4	Build Debug APK - <code>./gradlew assembleDebug</code>
5	Run unit tests - <code>./gradlew testDebugUnitTest</code>
6	Upload APK artifact (14-day retention)
7	Upload test result reports (14-day retention)

This keeps the main branch from breaking and makes sure all unit tests pass before any merge.

7.3 Unit Testing

Unit tests are present and merged into master via PR #1 (feature/unit-tests):

- CreatureTest - creature entity validation (stats, rarity bounds)
- GoopTypeTest - type system logic (fusion rules, getFusionResult correctness)
- PlayerStatsTest - player stats calculations (level formula, streak edge cases)

Tests run automatically in CI on every push.

7.4 Code Structure & Style

- Modular package structure: data/database/, data/repository/, ui/ar/, ui/map/, ui/collection/, ui/achievements/
- Separation of concerns: DAOs handle queries only; Repository handles business logic; Activities handle UI only
- Kotlin idioms: data classes, when expressions, extension functions, coroutines, Flow
- Code follows the Android Kotlin Style Guide (<https://developer.android.com/kotlin/style-guide>)
- KDoc and inline comments on key classes and algorithms

8. Third-Party Libraries & Assets

All third-party libraries are listed below as required. No library was used to implement the core classification logic of the CV algorithm.

Library	Purpose
androidx.room (runtime + ktx + compiler/KSP)	SQLite ORM - all data persistence
androidx.camera.camera2	Camera hardware access
androidx.camera.lifecycle	Lifecycle-aware camera binding
androidx.camera.view	PreviewView UI component
com.google.android.gms:play-services-location	Fused GPS location provider
kotlinx.coroutines.android	Async / non-blocking operations
androidx.lifecycle.viewmodel.ktx	Lifecycle-aware ViewModel scope

Library	Purpose
androidx.lifecycle.livedata.ktx	LiveData
androidx.lifecycle.runtime.ktx	lifecycleScope coroutine support
androidx.recyclerview	Scrollable list and grid displays
com.google.android.material	Material Design UI components
androidx.cardview	Card layout containers
androidx.navigation.fragment.ktx	Fragment navigation
junit	Unit testing framework
androidx.test.espresso.core	UI instrumented testing
androidx.test.ext:junit	Android JUnit test runner

Build Configuration: Android API 24 (min) - 34 (target/compile), JVM target 17, KSP for Room code generation.

Assets: All creature names, types, and game designs are original. rawable assets are AI-generated, created for this project

9. AI Usage Declaration

AI tools were used during this project as declared below:

Phase	Tool	How It Was Used
Design	Claude (Anthropic)	Brainstorming the colour-detection approach for AR-style spawning without ARCore
UI / Asset Design	Claude (Anthropic)	Generating creature names, Goopdex descriptions, colour palette suggestions, and app UI layout advice
Implementation	Claude (Anthropic)	Assistance with Room DAO query syntax, CameraX ImageAnalysis setup, coroutine patterns
Debugging	Claude (Anthropic)	Diagnosing Room database migration version conflicts and camera thread management issues

Phase	Tool	How It Was Used
Testing	Claude (Anthropic)	Writing unit test cases for GoopType fusion logic, player stats level formula, and streak edge cases
Report writing	Claude Code (Anthropic)	Structuring and drafting this report

Prompts used:

Design phase:

- "I want to build an AR creature spawning mechanic without using ARCore. Can I use the camera to detect colours in real time and use that to trigger spawns?"
- "What's a good way to structure a merge-to-evolve system like Dragon City in Android?"

“can you suggest 19 creature names for my game, based on 5 types: water fire nature electric and shadow, some should have 3 evolution stages like droplet → aqua → tsunami, need the names to sound cool”

“how should i design the catch mechanic, should success be based on rarity or should the player need to tap at the right time, also how many attempts before the creature escapes”

“what color palette should i use for a dark themed android mobile game where the creatures are glowing coloured blobs, i want it to look clean but still feel like a game”

Implementation phase:

- "My Room DAO isn't returning results when I join two tables. Here is my query:.... What is wrong?"
- "How do I set up CameraX ImageAnalysis to process frames on a background thread without blocking the UI?"
- "How do I use Kotlin Flow with Room so my RecyclerView updates automatically when the database changes?"
- "What's the correct way to use lifecycleScope and Dispatchers.IO together for database calls?"

“how do i request camera and location permissions at runtime in android, i already added them to the manifest but the app crashes on devices running android 6 and above”

“my recyclerview keeps showing the wrong creature image when i scroll, like the images are swapping around, how do i stop this”

“i want to let the player nickname their creature, how do i save that to the room database and update the display in the collection screen straight away”

“how do i make a custom view that draws a glowing circle on android canvas, i want a blur effect around the creature shape”

UI / Asset Design phase:

“give me pokedex style descriptions for these 19 creatures i made up, keep them short like 2 sentences, they should match the creature’s element type”

“how do i make a gradient header background in android using xml drawables, i want it to go from dark purple at the top to dark blue at the bottom”

“generate icon descriptions for nav bar buttons for a creature collecting game: scan, collection, map, i need them to feel game-like not generic”

Debugging phase:

- "I'm getting a Room migration error after adding a new column. My schema version is still 1. How do I fix this?"
- "My camera analyzer is running on the main thread and causing jank. How do I move it to a dedicated ExecutorService?"

“my evolution screen is always empty even when i have 3 of the same creature in my collection, the list just shows nothing, i think the filter is wrong but i cant figure out why”

“my login streak keeps resetting to 1 every time the app opens even if i opened it yesterday, the calendar comparison logic seems off”

Testing phase:

“how do i write a junit unit test for a kotlin function that takes two GoopType enum values and returns a fusion result, i want to test all 4 valid combos and some invalid ones”

“what should i be testing for a PlayerStats class, specifically the level calculation and the streak logic when the user opens the app 2 days in a row vs skipping a day”

Report writing phase:

- "Can you help me structure a technical report section explaining how my colour detection CV pipeline works?"
- "How should I document my database schema in a report - as a table or ER diagram?"

“can you help me write the CI/CD section of my report, we used github actions to build the apk and run tests on every push, how should i explain this in a technical report”

All AI suggestions were reviewed and tested by the team before being used. Final decisions and code were done by us.

10. Results & Evaluation

Features Delivered

Feature	Status
Camera colour detection (CV pipeline)	Complete
Colour to GoopType classification	Complete
Temporal hold filter (2 seconds)	Complete
Tap-to-catch with rarity probability	Complete
Creature escape timer (6 seconds)	Complete
Creature collection with filtering	Complete
Merge-to-evolve (3x same creature)	Complete
Fusion system (2-type hybrid, 4 recipes)	Complete

Feature	Status
GPS habitat mapping	Complete
Geohashing for deterministic habitat zones	Complete
Daily challenges with midnight expiration	Complete
Login streak system (cross-year aware)	Complete
Achievement system (10 achievements, 5 categories)	Complete
Room database (6 tables, seeded on first launch)	Complete
GitHub Actions CI/CD pipeline	Complete
Unit tests (3 test classes)	Complete

Mobile Features Summary (Requirement: 3+)

Feature	API / Component
Camera	CameraX ImageAnalysis
GPS / Location	Fused Location Provider
Database	Room ORM (SQLite)
Canvas Graphics	Android Canvas API
Animation	ValueAnimator, BlurMaskFilter
Multi-threading	Kotlin Coroutines + ExecutorService

6 mobile features total

Known Limitations

- Colour detection is sensitive to lighting conditions - very dark or very bright environments may not trigger spawns reliably
- Habitat zone positions are randomly seeded each session - zones are not spatially persistent across app restarts
- All data is stored locally; no cloud sync or remote leaderboard is implemented

Future Work

- Persist habitat zone seeds to the database so zones remain consistent across app restarts
- Implement cloud sync via Firebase to enable cross-device progress and a global leaderboard
- Expand the fusion recipe system with more type combinations and rare hybrid creatures
- Add push notifications for daily challenge reminders and login streak alerts
- Introduce a trading system allowing players to exchange creatures with nearby players via Bluetooth or NFC
- Implement a turn-based battle system allowing players to pit their Goops against each other or against AI-controlled encounters

Reflections

What went well:

- The colour-based CV pipeline proved more effective than initially expected - the 2-second temporal hold filter significantly reduced false spawns and made the mechanic feel responsive and intentional
- Using Room with Kotlin Flow meant the UI updated automatically without manual polling, which simplified a lot of the collection and home screen logic
- The Repository pattern kept the codebase clean and made debugging easier since all data logic was centralised in one place

What was harder than expected:

- Colour detection is inherently sensitive to lighting - achieving consistent spawns across different real-world environments required significant tuning of the saturation filter and RGB thresholds

- Room database migrations caused repeated issues during development when new columns were added, requiring careful versioning to avoid data loss on reinstall
- The EvolutionActivity initially used an XP-based filter that never surfaced any creatures, identifying and fixing this bug required tracing the filter logic back through the repository layer

What we would do differently:

- We would persist habitat zone seeds to the database earlier in development, as the current session-based random seeding means zones reset on every app restart, which reduces the sense of a persistent game world.

11. Links

Resource	URL
Source Code (GitHub)	<i>(paste GitHub repo link)</i>
App Demo Video	<i>(paste demo video link)</i>
Presentation Video	<i>(paste presentation video link)</i>